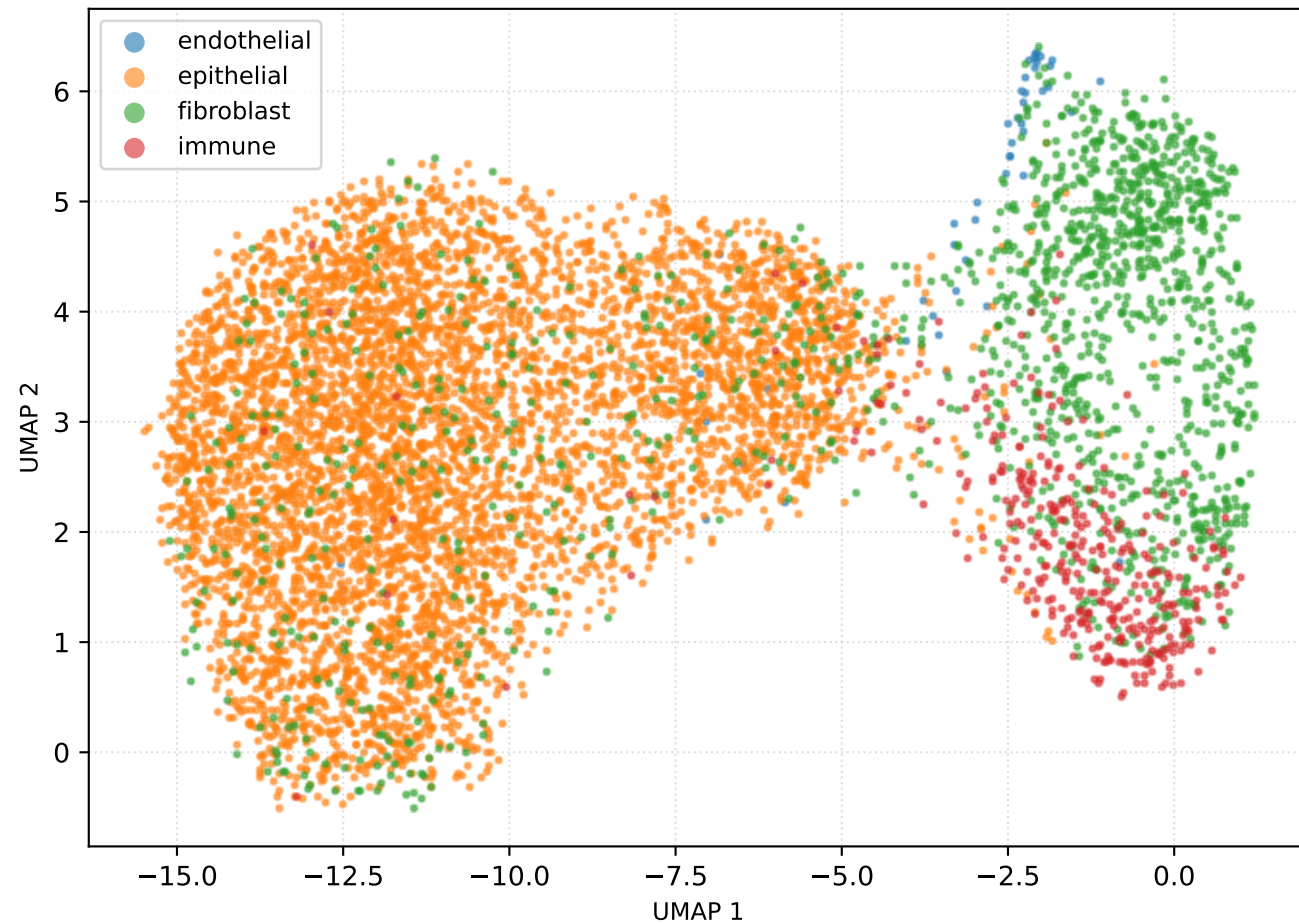


**Cell-Type Structure Preservation: Feature Comparison across Three Paradigms**  
[scGPT Zero-Shot: sil=0.4852 (ratio 0.92) | PCA-50: sil=0.1476 (ratio 1.00) | BiLSTM-50: sil=0.4287 (ratio 2.91)]

**PCA-50 Baseline (Raw Expression)**  
**Silhouette: 0.1476 | Ratio vs PCA-50: 1.00 (Reference)**



**BiLSTM-50 Spatial Neighbor Extractor**  
**Silhouette: 0.4287 | Ratio vs PCA-50: 2.9051**

